

UNIT III NETWORK LAYER

Course Objective - COB1: To learn the functions of network layer and the various routing protocols.

Course Outcome - CO1: Analyze and design routing algorithms

Bloom's Taxonomy Level: K1, K2, K3, K4, K5, K6

Part A

S.No.	Question & Answer	BTL*
1	State the function of the Network layer in the OSI model. The Network layer is the third layer of the OSI model responsible for carrying data packets from source to destination across multiple networks, using logical (IP) addressing and routing.	K1
2	Mention any two services provided by the Network layer. (i) Packetizing – encapsulating payload into network layer packets. (ii) Routing – finding the best path for packet delivery.	K1
3	Specify the address sizes used in IPv4 and IPv6. IPv4 → 32 bits (4 bytes) IPv6 → 128 bits (16 bytes)	K1
4	List any two types of IPv6 addresses. (i) Unicast (one-to-one) (ii) Multicast (one-to-many)	K1
5	Give the definitions of subnetting and supernetting. Subnetting: Dividing a network into smaller sub-networks. Supernetting: Combining multiple smaller networks into a larger one.	K1
6	State what RIP stands for and define it. Routing Information Protocol (RIP) is a simple intradomain distance vector routing protocol used inside an Autonomous System (AS), in which each router periodically shares its knowledge of the entire AS with its neighbors.	K1
7	Mention the maximum hop count allowed in RIP. The maximum hop count in RIP is 15; a hop count of 16 is considered infinite (unreachable).	K1
8	Name the two major interdomain routing protocols used in the Internet. (i) Exterior Gateway Protocol (EGP) (ii) Border Gateway Protocol (BGP)	K1
9	Differentiate between static routing and dynamic routing.	K2

	Static routing: Routes are manually configured by the administrator and do not change automatically. Dynamic routing: Routing protocols automatically discover and update paths in response to network changes.	
10	Compare Intradomain and Interdomain routing protocols. Intradomain routing operates inside a single Autonomous System (e.g., RIP, OSPF) and uses distance vector or link state methods, while Interdomain routing operates between different Autonomous Systems (e.g., BGP) and uses path vector routing.	K2
11	Illustrate the concept of an Autonomous System (AS) with an example. An Autonomous System (AS) is a group of networks and routers under the authority of a single administration. Routing inside an AS is intradomain routing, while routing between different ASs is interdomain routing.	K2
12	Differentiate between Distance Vector Routing and Link State Routing. In Distance Vector Routing (DVR), each router periodically shares its knowledge about the entire network only with its neighbors. In Link State Routing (LSR), each router shares its knowledge of its immediate neighborhood with all routers in the network whenever there is a change.	K2
13	Outline the advantages of OSPF over other routing protocols. OSPF offers authentication, support for hierarchical routing domains, multiple equal-cost paths, and integrated support for both unicast and multicast routing.	K2
14	Describe the three functions performed by BGP within an Autonomous System. BGP allows each AS to: (i) obtain subnet reachability information from neighboring ASs, (ii) propagate this information to internal routers, and (iii) determine good routes to subnets based on reachability information and AS policy.	K2
15	Discuss the limitations of EGP that led to its replacement by BGP. EGP was designed for a tree-like Internet topology, so it only allowed autonomous systems to be connected as parents and children rather than as peers, limiting its use in a general mesh topology. It was later replaced by BGP.	K2
16	Illustrate multicast addressing and specify its IP address range in IPv4 and IPv6. A multicast address identifies a dynamic group of hosts; any host that is a member of the group receives a copy of packets sent to that address. In IPv4, multicast addresses use Class D (224.0.0.0 to 239.255.255.255); in IPv6, they use addresses with the prefix 1111 1111.	K2
17	Summarize the contents of a Link State Packet (LSP).	K2

	An LSP contains: the ID of the node, the list of directly connected neighbors and cost to each, a sequence number, and a time-to-live value. Every node floods its LSP over each of its directly connected links.	
18	Compare the working of PIM-Dense Mode and PIM-Sparse Mode. PIM-DM assumes receivers are densely distributed across the network, so it floods multicast traffic everywhere and prunes unwanted branches. PIM-SM assumes receivers are sparsely distributed, so it uses a Rendezvous Point (RP) and only forwards traffic to routers that explicitly join the group.	K2
19	Explain the advantages of hierarchical network design. (i) Scalability – supports growth to large networks by organizing them into levels. (ii) Simplified management and troubleshooting, since faults and routing information can be localized within a level.	K2
20	Interpret the role of Reverse Path Forwarding (RPF) in multicast routing. RPF is a loop-prevention technique used in multicast routing (e.g., DVMRP) in which a router forwards a multicast packet only if it arrives on the interface that lies on the shortest path back to the source; otherwise the packet is discarded.	K2

Part B

S.No.	Question	BTL*
1	Apply subnetting principles to divide a given Class B IPv4 network into equal-sized subnets, specifying the subnet mask and host range for each subnet. (16 Marks)	K3
2	Demonstrate the working of the Distance Vector Routing (RIP) algorithm using a suitable network example, showing routing table updates at each iteration. (16 Marks)	K3
3	Summarize the working of Link State Routing with reference to OSPF, including the flooding process and Dijkstra's shortest-path computation. (16 Marks)	K4
4	Compare and contrast the addressing structures of IPv4 and IPv6, highlighting header formats and the advantages of IPv6 over IPv4. (16 Marks)	K2
5	Discuss why BGP replaced EGP as the standard interdomain routing protocol, evaluating its design and performance. (16 Marks)	K2
6	Apply DVMRP and PIM multicast routing protocols to suitable dense and sparse network scenarios and demonstrate how packets are delivered efficiently. (16 Marks)	K3
7	Analyze a hierarchical routing scheme for a large Autonomous System and explain how its structure improves scalability and simplifies route management. (16 Marks)	K4
8	Develop a multicast routing strategy for a network with sparsely distributed receivers, and justify your design choices with reference to PIM-SM and Reverse Path Forwarding. (16 Marks)	K6